

17/6/2022

$$D9) \quad T(n) = 8T(n/3) + \underbrace{2^n}_{\text{pink box}}$$

$$n^{\log_b a} = n^{\log_3 8} \approx n^{1.89}$$

CASO 3

$$2^n = \Omega(n^{\log_3 8 + \varepsilon})$$

$$\underbrace{a f(n/b) \leq c \cdot f(n)}_{\text{yellow box}} \\ 8 \cdot 2^{n/3} \leq c \cdot 2^n$$

$$\underbrace{\frac{8 \cdot 2^{n/3}}{2^n} \leq c}_{\text{yellow box}}$$

$$\underbrace{8 \cdot \left(\frac{1}{4}\right)^{n/3}}_{\text{yellow box}} \leq c \quad \square$$

$$\boxed{T(n) = \Theta(2^n)} \quad \leftarrow$$

Algoritmo (n)

{ $c = 0$; $a = n$

while $a > 1$

{ $b = 1$ ←

while $b \leq a^2$

{ $c = c + 1$ ←

$b = 2 \cdot b$ ←

$a = a/2$ ←

}

return c ←

}

$$\sum_{a=1}^{\log n} 2 \cdot \log \frac{n}{2^i}$$

$$n, \frac{n}{2}, \frac{n}{4}, \dots, \frac{n}{2^i}$$

$$\frac{n}{2^i} = 1$$

$$\log(n) = \log(2^i)$$

$$\Rightarrow i = \log n$$

$$1, 2, 4, 8, \dots, 2^j$$

$$\log(2^j) = \log(a^2)$$

$$\Rightarrow 2 \cdot \log a$$

$$O(\log^2 n)$$

5/7/2021

D7) $\underbrace{O(n^2)}_{\text{iterativo}}$

$$T(n) = \begin{cases} n & n \leq 1 \\ aT\left(\frac{n}{4}\right) + 1 & \underline{n > 1} \end{cases}$$

$$T(n) = aT\left(\frac{n}{4}\right) + 1$$

$$\underbrace{1} \quad \underbrace{n^{\log_4 a}} \quad \underbrace{a=16}$$

$$n^{\log_4 16} = \underbrace{n^2}_{\text{CASO 1}}$$

$$\Theta\left(n^{\log_4 16}\right) = \Theta(n^2)$$

$a < 16 \Rightarrow$ ricorsivo

$a = 16 \Rightarrow$ equivalenti

$a > 16 \Rightarrow$ iterativo

D8)

for ($i = 0; i < n; i++$)

for ($j = 0; j < n; j++$)
 $a[j] = \text{randomValue}(i)$ | n

Sort(a) $\longrightarrow O(n \cdot \log n)$

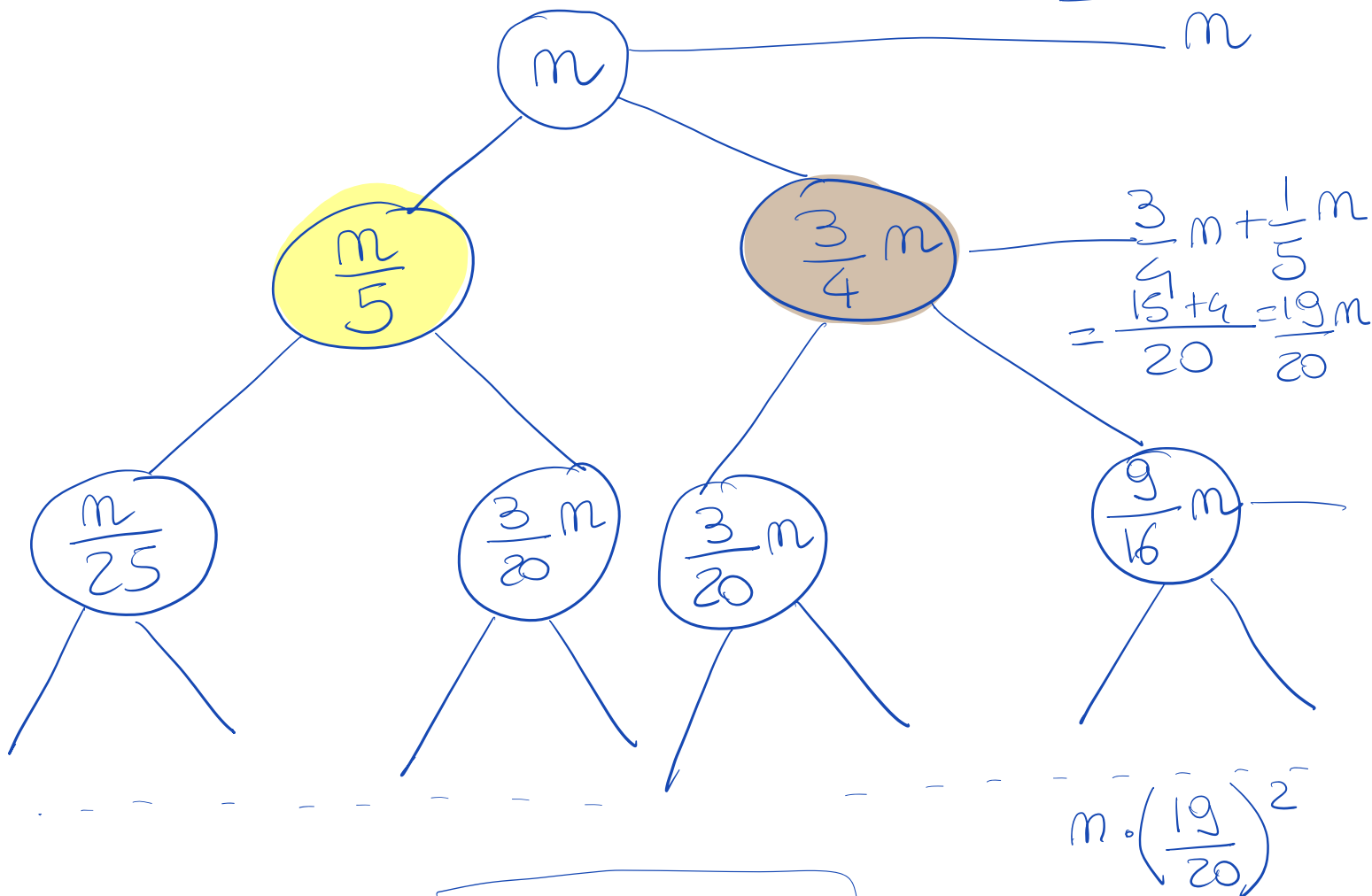
$$(n + n \cdot \log n) \cdot n$$

$$O(\cancel{n^2} + n^2 \cdot \log n)$$

$$O(n^2 \cdot \log n)$$

29/7/2022

D9) $T(n) = T\left(\frac{n}{5}\right) + T\left(\frac{3n}{4}\right) + \Theta(n)$



$$T(n) \leq \sum_{k=0}^{\log n} n \left(\frac{19}{20}\right)^k$$

$$= n \cdot \left(\frac{19}{20}\right)^{\log n} = O(n)$$

$$\left(\frac{3}{4}\right)^{\log n} \cdot n = 1$$

$$m = \left(\frac{4}{3}\right)^J$$

$$\frac{3}{4} \log m = J$$

Procedura **F(n)**

```

{  if  $n \leq 2$  then return  $n$ 
    else
         $i = \text{arrotondo\_dif}(n/3)$ 
         $j = \text{arrotondo\_ecc}(n/3)$ 
         $x = F(i)$ 
         $y = F(j)$ 
         $p = P(n)$ 
        return  $px + y$ 
}

```

Procedura **P(n)**

```

{  if  $n = 0$  then return 1
    else
         $k = \text{arrotondo\_difetto}(n/2)$ 
         $x = P(k)$ 
        if  $n$  è pari then return  $x \cdot x$ 
        else return  $2 \cdot x \cdot x$ 
}

```

$$T_P(m) = 1 \cdot T_P(m/2) + C$$

$$\Theta(\log m)$$

$$T_F(m) = 2 \cdot T_F\left(\frac{m}{3}\right) + \log m + C \quad \leftarrow$$

$$m^{\log_a b} = m^{\log_3 2}$$

$$m^{0.6}$$

$$\log m$$

$$\log m$$

Caso 1

$$O(m^{0.6-\epsilon})$$

$$\Theta(m^{\log_3 2})$$

25/2/2022

$$D7) \quad T(n) = T(n-1) + \log n$$

$$T(n) = O(n \cdot \log n)$$

$$T(n) \leq c \cdot n \cdot \log n \quad \Leftarrow$$

$$T(n-1) \leq c(n-1) \cdot \log(n-1) \quad \begin{array}{l} \text{IPOTESI} \\ \text{VERA} \end{array}$$

$$T(n) \leq \underbrace{c(n-1) \cdot \log(n-1)} + \log n \leq c \cdot n \cdot \log n$$

$$\leq \underbrace{c(n-1) \cdot \log(n)} + \log n \leq c \cdot n \cdot \log n$$

$$\leq \cancel{c n \cdot \log n} - c \cdot \log n + \log n \leq \cancel{c \cdot n \cdot \log n}$$

$$(-c+1) \cdot \log n \leq 0$$

$$\Rightarrow \boxed{c > 1}$$

D8)

Algoritmo (A, i, j)

A è array di numeri interi ordinati e distinti.

{ if $i \leq j$ then

{ $m = \lfloor (i+j)/2 \rfloor$

T.left = Algoritmo (A, i, m-1)

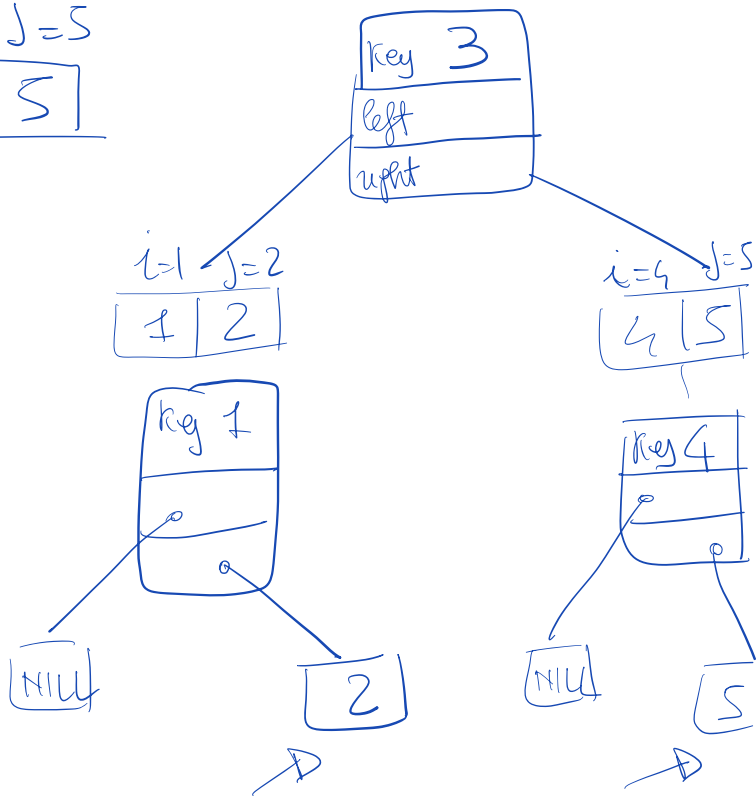
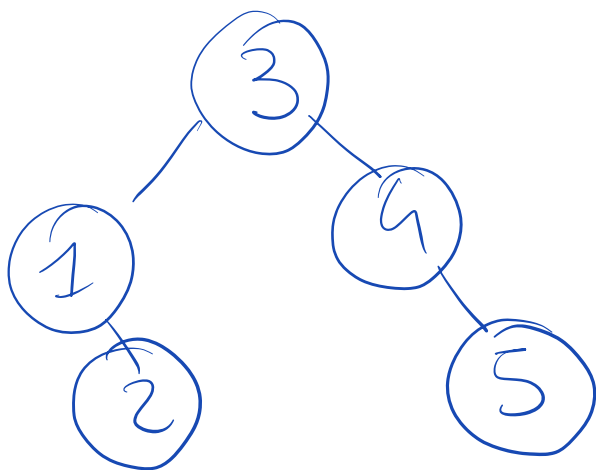
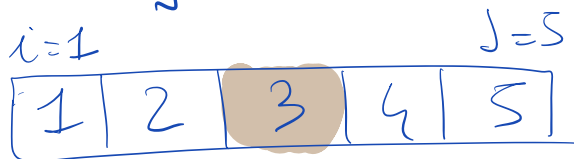
T.right = Algoritmo (A, m+1, j)

T.Key = A[m]

} return T

} else

return nil



$$T(n) = 2T(n/2) + C \Rightarrow \Theta(\log n)$$

D2)

